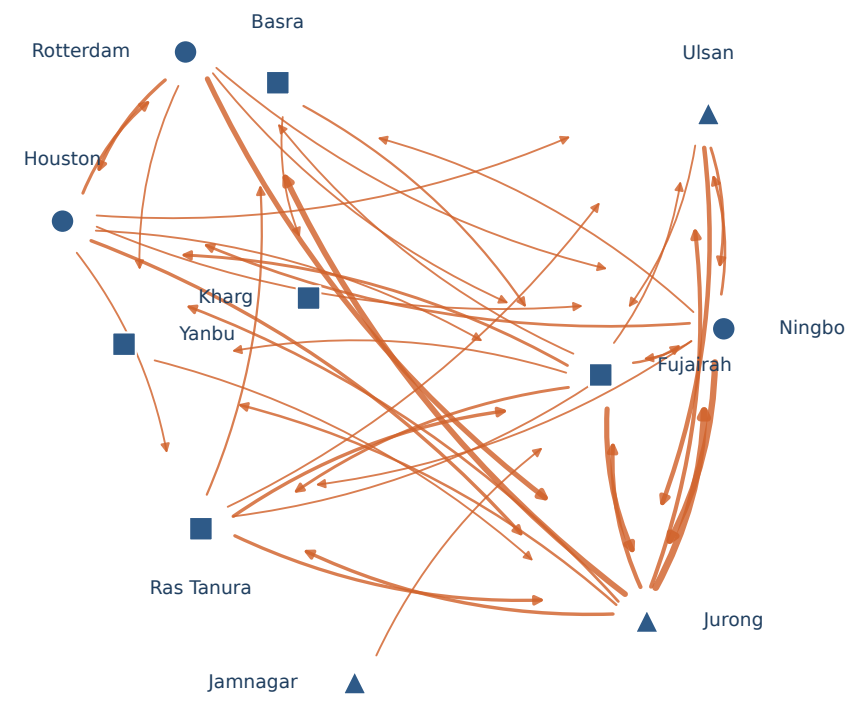


Two edge classes in the 17-node shipping graph

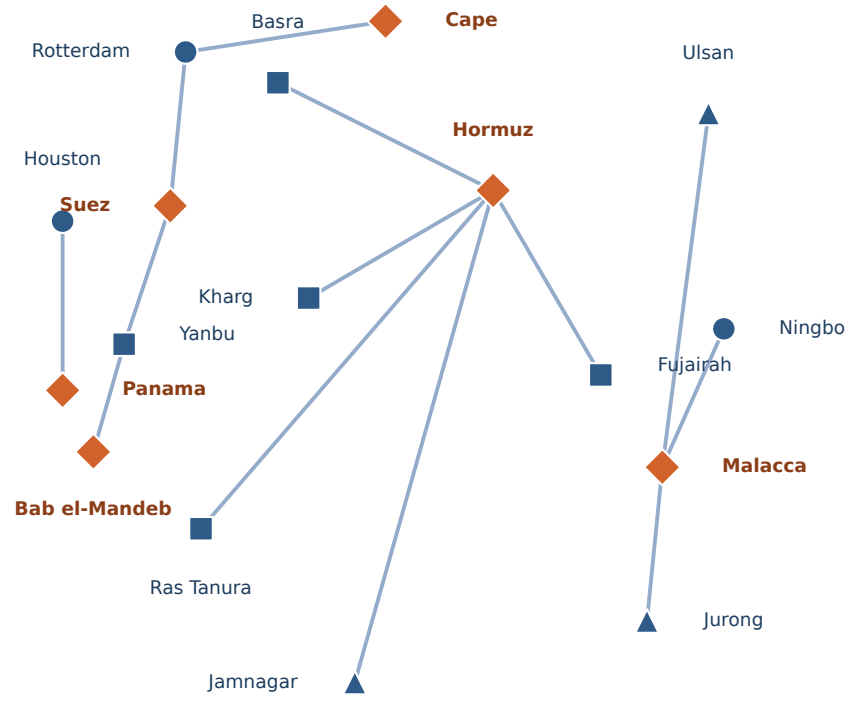
(a) Weekly voyage graph · directed and weighted

Illustrative median-density week. Arrows indicate direction; width indicates weekly voyage count.



(b) Fixed corridor graph · undirected and unweighted

Thirteen ex-ante AOI-chokepoint links, identical in every week.



● AOI · port ■ AOI · terminal ▲ AOI · refinery ◆ Chokepoint — Directed voyage edge — Undirected corridor edge

Entering the GAT: the two classes are stacked, then symmetrised and self-looped. Direction and edge type are not retained.

Across the full sample, the weekly voyage graphs comprise 96 unique directed lanes and 106,992 voyages. Panel (a) is the week ending 1 July 2022 (43 lanes = sample median of 43).